Experiment-3 *Projectile Motion*

In this part of the experiment, the puck is projected horizontally with an initial velocity. By the dots produced on the data sheet, we will investigate the motion of the puck along the horizontal and vertical axes independently.

Theory

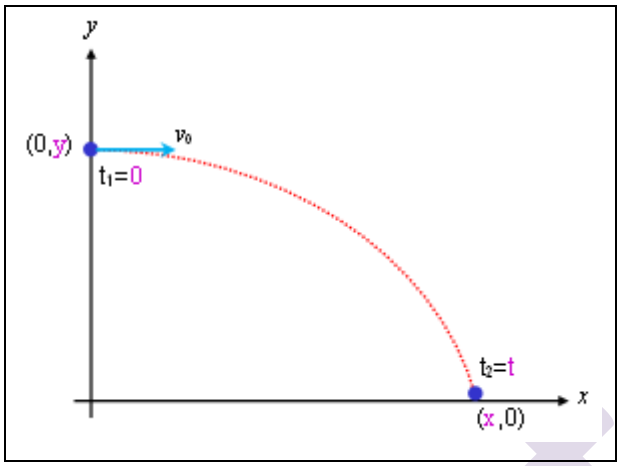


Figure-8: Projectile motion of an object projected horizontally. Trajectory of the object fired horizontally at time $t_1=0$, with the initial speed v_0, from a point at (0; h). The dashed line represents the path of the object.

The other type of the motion in this experiment is the horizontally **projected motion**. Projectile motion is the **two-dimensional** motion of an object under the influence of Earth's gravitational acceleration (***g***). A **projectile** is any object that is given an initial velocity and then follows a path determined entirely by the effects of gravitational acceleration. The **path** followed by a projectile is called its **trajectory**. We consider only its motion *after* it has been projected, and *before* it lands or is caught.

As seen in the **Figure-(8)**, the position of such an object at any given time **t** is given by a set of coordinates x and y which vary with respect to time and represent the horizontal and vertical coordinates, **respectively**.

One of the two components of the **velocity vector** is parallel to the **horizontal** x-axis and the other is parallel to the **vertical**, or y-axis:

$$\vec{v} = v_x\hat{i} + v_y\hat{j} \tag{10}$$

The motion of the object in the horizontal x-direction is a straight line motion with **constant velocity**. In other words, the x-component of the velocity of the object (v_x) will be constant.

However, the acceleration only acts along the vertical direction. The x-component of acceleration is **zero** and y-component is **constant**. This means that only the vertical component of the velocity (v_y) will change with respect to time and the horizontal component of the velocity will be **constant**. Therefore, we can analyze projectile (**two-dimensional**) motion as a combination of horizontal motion with *constant* velocity and the vertical motion with *constant* acceleration.

We can express the vector relationships for the projectile's acceleration by separate equations for the horizontal and vertical components. The components of **the acceleration vector** $\vec{a}$ are:

$$a_x = 0 \tag{11}$$

$$a_y = a \text{ (Constant)} \tag{12}$$

Since there is no acceleration in the horizontal x-direction, the horizontal component of the particle's velocity remains *constant* from its initial value throughout the motion.

In vector form, the acceleration can be expressed as the form:

$$\vec{a} = a_y\hat{j} \tag{13}$$

The position of the object at a given time is given by:

$$\vec{r} = x\hat{i} + y\hat{j} \tag{14}$$

For the two-dimensional (x, y) motion, we can separate acceleration, displacement and velocity in both x and y coordinate directions by the general equations:

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \quad (15)$$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \quad (16)$$

We can write the kinematic equations for constant acceleration (a) in two-dimensional motion as:

x-Component (Horizontal)	y-Component (Horizontal)
$v_x = v_{0x} + a_x t$	$v_y = v_{0y} + a_y t$
$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$	$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$

Here,

v_x : Horizontal component of velocity,

v_{0x} : **Initial** velocity in the horizontal (x)-direction,

a_x : **Acceleration** in the horizontal (x)-direction,

t : Time for the motion,

x : Horizontal displacement,

x_0 : **Initial** position in the horizontal (x) direction,

v_y : Vertical component of the velocity,

v_{0y} : **Initial** velocity in the vertical (y) direction,

a_y : **Acceleration** in the vertical (y) direction,

y : Vertical displacement,

y_0 : **Initial** position in the vertical (y) direction,

In order to model two-dimensional projectile motion, a metal puck will be set in motion on an inclined **air table**. When the air compressor is switched on, the air is supplied to down the tube under the puck so that it moves with a given initial velocity on the frictionless plane.

Suppose that at time t=0, the particle (puck) is at the point (x_0, y_0) and at this time its velocity components have the initial v_{0x} and v_{0y}. Since the velocity of **horizontal motion** in the projectile motion is constant, we find:

$$a_x = 0 \quad (17)$$

$$\boxed{v_x = v_{0x}} \quad \textbf{(Experimental)} \quad (18)$$

$$x = x_0 + v_x t \quad (19)$$

We can choose the axes of our coordinate system at the point of release of the puck from the shooter at t=0 and take the **downward** direction to be **positive** such that:

$$x_0 = y_0 = 0 \quad (20)$$

This is the origin in the coordinate system associated with the initial position of the puck at t=0. Now, using this relationship in the Equation-(**19**), we will find the equation of motion along the x-axis as:

$$\boxed{x = v_x t} \quad \textbf{(Experimental)} \quad (21)$$

The horizontal displacement for the projectile motion of a small ball projected horizontally is determined by using Equation-(21). If the **horizontal** displacement (x) of the motion and time of flight (t) are measured, then we can find the initial velocity of the object.

For the motion along the y-axis, the velocity (v_y) and **vertical displacement** (y) at the later time (t) become:

$$v_y = at \quad (22)$$

$$\boxed{y = \frac{1}{2}at^2} \quad \textbf{(Experimental)} \quad (23)$$

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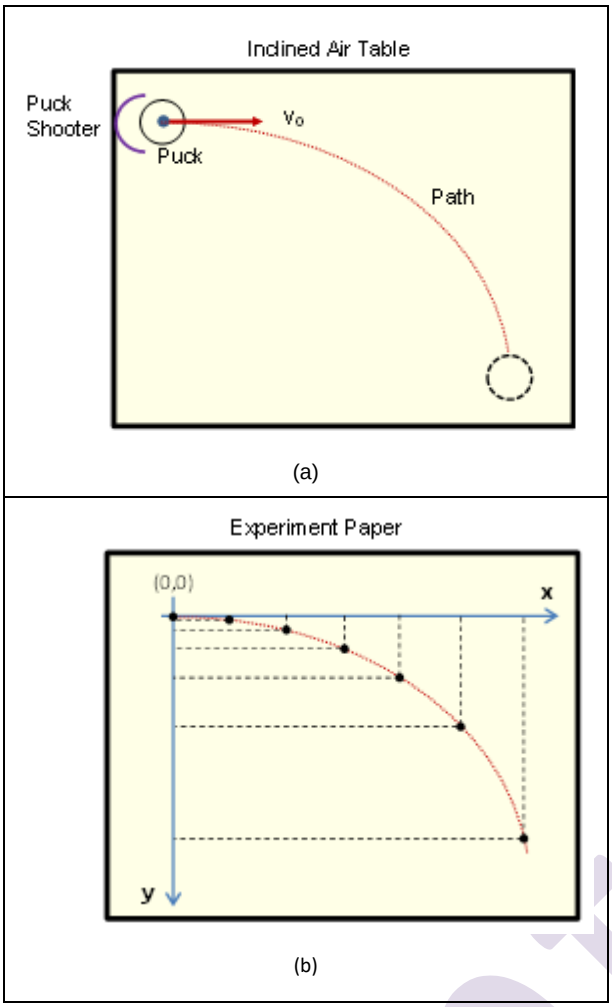


Figure-9: Schematic diagram of the horizontally projected motion of the puck on an inclined air table (a). The data points produced by the puck on the data sheet (b). The ***x*-component** of the velocity of the puck is **constant**. As for motion along the ***y*-axis**, note that the *distance* between the ***y*-projections** of the puck increases with time.

The vertical y-component of the velocity is obtained experimentally and used to determine the acceleration along the y-axis. The projectile motion has a constant horizontal *velocity* and a constant vertical (downward) *acceleration* due to gravity. The vertical distance (y) caused by the change in the velocity is given by Equation-(23).

The dots produced on the data sheet will look like the figure as shown in the Figure-(9). Here, note that the intervals between the dots of the ***x*-projections** in the horizontal direction are equal.

In the projectile motion, the *horizontal* motion and the *vertical* motion are independent of each other, neither motion affects the other. This feature allows you to separate a problem involving two-dimensional motion into two easier one-dimensional problems; one for the horizontal motion (with zero acceleration) and one for the vertical motion (with constant downward acceleration).

We choose our time interval to **begin** (t=0) just as the puck leaves the puck shooter at the position x_0=0, y_0=0. In the horizontal x-direction, the acceleration a_x=0 so the velocity of the puck is constant. In the vertical direction, the distance between the y-projection of the puck increases with time. This situation is identical to the one we have studied earlier, in the case of motion with acceleration (inclined air table). Finally, if you analyze the projectile motion, you can make the following important conclusions:

- The **horizontal** component (***x*-axis**) of a projectile's velocity is constant. (So, the horizontal component of acceleration, in other words, is zero).
- The projectile motion will have a constant **downward** (*y*-axis) **acceleration** due to gravity as seen in the Figure-(**9b**).

3.1. *Experimental Procedures*

In this part of the experiment, you will perform measurements on the motion of a horizontally projected puck on the inclined air table and the motion in two dimensions will be analyzed.

1. First, level off the air table and then bring it into an inclined position with respect to the horizontal by using the block to give an inclination angle:
   - $\theta = 9^0$.

2. Turn on the spark timer and adjust the frequency of the spark timer to:
   - f=20Hz.

3. Keep **one** of the pucks stationary on a folded piece of data sheet paper and carbon paper at the lower corner of the plane.

4. Attach the **shooter** (projectile launcher) to the left side of the inclined plane of the air table about 10cm from the upper edge and **adjust** the shooting angle to 0^0 (zero degrees) **to give horizontal shooting**.

5. First, activate only the compressor switch (P), place the puck into the shooter and make a few test shootings to adjust the best tension of the rubber belt of the shooter to give a convenient trajectory.

6. Now, activate the compressor switch (P) and **place** the puck into the adjusted shooter, then **release** the puck from the shooter **while simultaneously** activating the spark timer (S).

To avoid electrical shock, do not touch the conducting portions of the pucks or the air table while the spark timer is on.

6.1. Stop pressing both pedals when the puck comes to the bottom of the data sheet.

6.2. You will denote these dots of the data points by the **trajectory-B**.

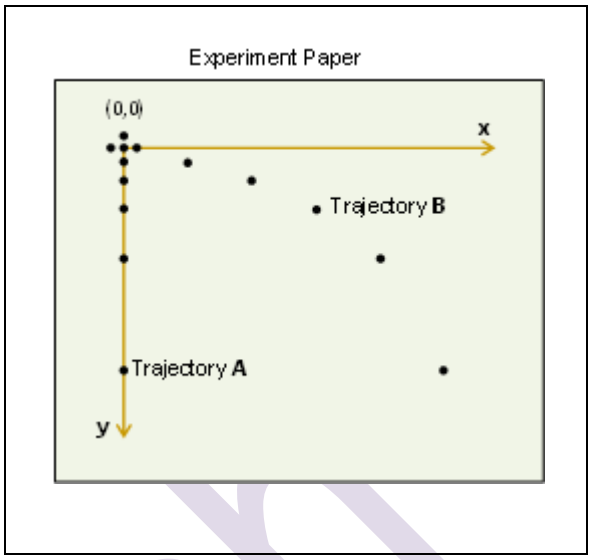


Figure-10: The dots as data points produced by the puck on the data sheet.

7. **On the same data sheet**, **place** the puck opposite to the shooter without tension of shooter (note that the puck must be outside the shooter).

7.1. Then activate both **compressor** and **spark timer** switches in the same time and then **release** the puck from the same initial height, without launching it on the inclined plane.

7.2. This trajectory will be a straight line and the dots will give **trajectory-A**.

8. **Turn off the spark-timer and then remove the pucks from the air table**.

9. Now, **remove the data sheet** to examine the dots of trajectories you have got. You must get the trajectories illustrated in Figure-(10). If the data points are inconvenient to analyze, repeat the experiment and get new data.

10. Select a clear dot on the path at x**=0** and **t=0** as the initial position of the motion.

10.1. Circle and number the data points starting from the first dot as **0, 1, 2, 3, 4, 5,…,10**.

10.2. Consider the downward trajectory as positive y-**axis** and horizontal trajectory as positive x-**axis**.

11. Now, you are ready to analyze projectile motion, horizontally and vertically.

    11.1. First, draw the x and y-axes for **trajectory-B** to determine the **initial position** of the projectile motion.
    11.2. This can be done by first drawing a line parallel to the **trajectory-A** that passes the first dot (**dot 0**) of the **trajectory-B**. This line will give the y**-axis** (take the positive y-direction to be downwards).
    11.3. Then, draw a horizontal line perpendicular to it from **dot 0** to get the x**-axis**.

12. After that, draw from each dot in **trajectory-B** normal to the x and y-axes to get the x and y**-projections** of these dots.

13. Measure the time of flight, t_f (the total time that elapsed during the motion of the puck) and the range x_R of the projectile (the horizontal distance traveled during the motion).

    13.1. Note that the time interval **between two dots** is given by "$1/f$" which is equal to 1/20 seconds.
    13.2. Calculate the projection velocity (*horizontal velocity*) $v_x=v_{0x}$ by using the **time of flight** (t_f) and total **horizontal distance** (x_R) in the Equation-(21).
    13.3. Record your data in the Table-(5).
    13.4. Note that in the horizontal (x) direction of an object projected horizontally, the initial velocity is horizontal and constant (that is, a_x=0).

14. Starting from dot "**0**" of the **trajectory-*B***, measure the distances of the y**-projections** of **10** data points (dots) with respect to the point (origin).

    14.1. Determine also the time (**t**) corresponding to each of these dots.
    14.2. Fill the measurements in the **trajectory-B** columns in the experimental data Table-(**6**).

15. Similarly, by starting from dot "**0**" of the **trajectory-*A***, **measure** the distances (locations) of the first **10** data points (dots) with the corresponding times along the y-axis.

    15.1. For **trajectory-A**, you can calculate the time (**t**) corresponding to each of the dots by using spark-timer frequency.
    15.2. Record your data values in the **trajectory-A** columns of the Table-(6).

10. **Plot** "y" versus "t^2" graphs for both trajectory-A and trajectory-B. Then, draw the best line for the each graph that fits your data points.

    10.1. By using the slope, find the accelerations:
    - a_A and,
    - a_B

    of the **vertical** motions for both trajectory-A and trajectory-B.

    15.3. **Note that** the **slope** of the x versus t^2 graph will be equal to: "$\frac{1}{2}a$".
    15.4. Compare these two accelerations of both trajectories and also compare them with the **acceleration** found in the previous experiment (Experiment-2).
    15.5. Show your calculations with the correct number of significant figures and **units** on your graph paper clearly.

Conclusions and Questions

Path-1 (straight line-trajectory-A):

- Is the **acceleration** constant for the motion along the y-axis?

Path-2 (curved line- trajectory-B):

- Is the **horizontal velocity** of a projected puck constant? Use your graph and data sheet for your explanation.
- If the **velocity** of horizontal motion in the projectile motion is constant, what is the **horizontal acceleration** of a projected puck?.
- Does the **vertical velocity** of the object increase downward in each time interval?.
- Is the **vertical acceleration** of projectile motion constant?.

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3.2. LABORATORY REPORT

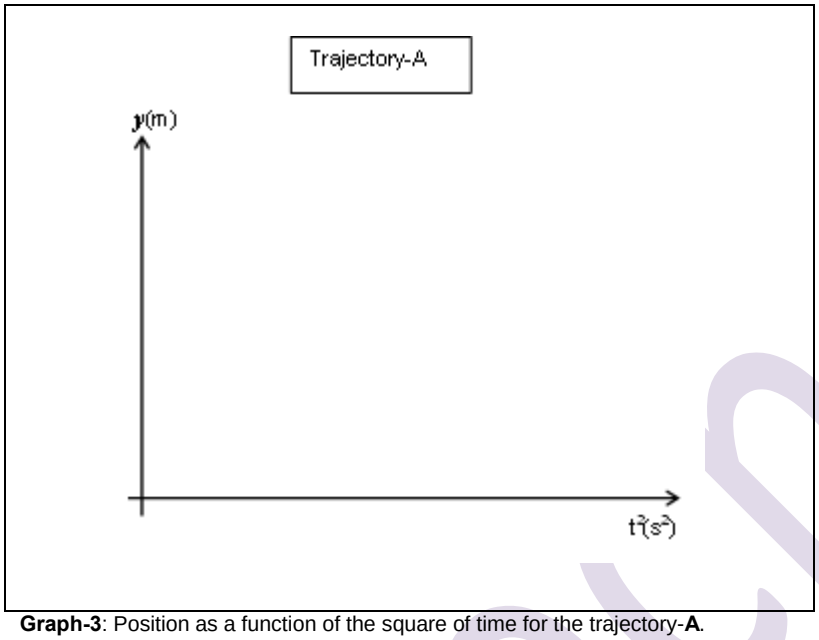
Projectile Motion

Table-5: Horizontal velocity of the **projectile motion**.

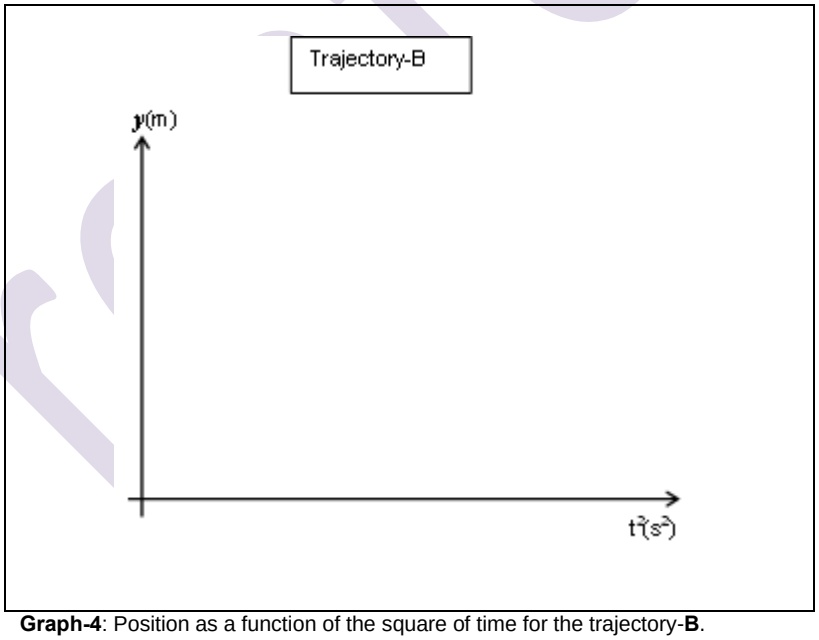
Frequency $f\,(Hz)$	Dot Number	Trajectory-*B* (x -Horizontal Component Motion)				
		$x(m)$	$t\,(\sec)$	$x_R\,(m)$	$t_f\,(\sec)$	$v_x\,(m/s)$
		(Measured)	(Measured)	(Calculated)	(Calculated)	(Calculated)
20	0	0	0	…..	…..	…..
	1					
	2					
	3					
	4					
	5					
	6					
	7					
	8					
	9					
	10					

Table-6: The measurements of accelerations for the trajectory-*A* and trajectory-*B*.

Dot Number	Trajectory-*A*		Trajectory-*B*		Trajectory-*A*	Trajectory-*B*
	(Vertical Motion)		(Vertical Motion)		(By the Slope)	(By the Slope)
	$y\,(m)$	$t\,(\sec)$	$y\,(m)$	$t\,(\sec)$	$a_A\,(m/s^2)$	$a_B\,(m/s^2)$
0	0	0	0	0	…..	…..
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						



Graph-3: Position as a function of the square of time for the trajectory-**A**.



Graph-4: Position as a function of the square of time for the trajectory-**B**.